

Literature-Implied Answer to the L^p Non- R -Schauder Basis Question

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Status

This is a lightweight literature-implied answer packet. The source question is answered affirmatively by combining a later general statement of Arnold–Le Merdy with the classical fact that $L^p([0, 1])$, $1 < p < \infty$, has an unconditional basis and is not Hilbert when $p \neq 2$.

Source Question

Source paper: Christoph Kriegler and Christian Le Merdy, *Tensor extension properties of $C(K)$ -representations and applications to unconditionality*, arXiv:0901.1025.

The question appears in Remark 5.6(2), page 22. After recalling that Kalton–Lancien imply that $L^p([0, 1])$, $1 < p \neq 2 < \infty$, admits a finite-dimensional Schauder decomposition which is not R -Schauder, the authors write:

“However the question whether $L^p([0, 1])$ admits a Schauder basis which is not R -Schauder, is apparently an open question.”

Supporting Answer

Supporting paper: Loris Arnold and Christian Le Merdy, *New counterexamples on Ritt operators, sectorial operators and R -boundedness*, arXiv:1812.11299.

On page 7, immediately after Corollary 0.4, Arnold–Le Merdy state:

“It follows from [4,6] that if X has an unconditional basis and X is not isomorphic to a Hilbert space, then X has a Schauder basis which is not R -Schauder.”

Here [4] is Fackler’s revisiting of the Kalton–Lancien theorem and [6] is Kalton–Lancien, *A solution to the problem of L^p -maximal regularity*.

Identification

For $1 < p < \infty$, the Haar system is an unconditional Schauder basis of $L^p([0, 1])$. For $p \neq 2$, $L^p([0, 1])$ is not isomorphic to a Hilbert space. Applying the Arnold–Le Merdy statement with

$$X = L^p([0, 1])$$

therefore gives a Schauder basis of $L^p([0, 1])$ which is not R -Schauder. This gives an affirmative answer to the full range asked in Remark 5.6(2) of arXiv:0901.1025.

Scope and Provenance

This is not a new proof from the run. The supporting authors do not appear to explicitly identify Remark 5.6(2) of arXiv:0901.1025 as the question being answered. The relation is an agent-identified implication from their general statement.

The source's broader applications to $C(K)$ -representations, H^∞ calculus, and unconditionality are not reclassified here; only the specific $L^p([0, 1])$ basis question in Remark 5.6(2) is cleared.

References

- [1] C. Kriegler and C. Le Merdy, *Tensor extension properties of $C(K)$ -representations and applications to unconditionality*, arXiv:0901.1025. <https://arxiv.org/abs/0901.1025>
- [2] L. Arnold and C. Le Merdy, *New counterexamples on Ritt operators, sectorial operators and R -boundedness*, arXiv:1812.11299. <https://arxiv.org/abs/1812.11299>
- [3] S. Fackler, *The Kalton–Lancien theorem revisited: maximal regularity does not extrapolate*, J. Funct. Anal. **266** (2014), 121–138.
- [4] N. J. Kalton and G. Lancien, *A solution to the problem of L^p -maximal regularity*, Math. Z. **235** (2000), 559–568.